

Eventual negative real-rootedness for fixed-rank paving matroid base polytopes and a growing-rank root-density theorem

Carptopus

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Abstract

Let M be a paving matroid and $h_M^*(x)$ the Ehrhart h^* -polynomial of its base polytope. We prove two complementary asymptotic results. First, for every fixed rank $r \geq 3$, all zeros of h_M^* are simple and strictly negative, uniformly over all sufficiently large n -element rank- r paving matroids; no connectivity assumption is imposed. Second, when the rank grows with the ground-set size, we give an explicit regime in which a connected paving matroid has $D - o(D)$ distinct strictly negative zeros, where $D = \deg h_M^*$. The second theorem assumes a mild upper bound on the sizes of effective hyperplanes and does not classify the remaining $o(D)$ zeros. The proof combines the paving hyperplane-correction formula with uniform phase grids, endpoint scaling limits, quantitative spectral gaps, and packing estimates for the hyperplanes. The two theorems concern different asymptotic regimes and neither implies the other.

This paper is accompanied by `proof-dossier.md` and its frozen 216-file SHA-256 closure. The dossier is an integral supplement: whenever a long uniform contour estimate is summarized in the main text, the cited dossier section supplies the complete inequalities, endpoint certificates, and quantifiers. The theorem statements and assembly arguments do not rely on historical workflow labels in those records.

1 Main results

Throughout, M is a loopless rank- r matroid on $[n]$, $P(M)$ is its base polytope in its intrinsic lattice, and

$$1 + \sum_{q \geq 1} |qP(M) \cap \mathbb{Z}^n| x^q = \frac{h_M^*(x)}{(1-x)^{\dim P(M)+1}}.$$

A matroid is **paving** if every circuit has size at least its rank. An effective hyperplane is a rank- $(r-1)$ hyperplane H with $|H| \geq r$. Distinct effective hyperplanes of a paving matroid intersect in at most $r-2$ elements. We write $\mathcal{H}(M)$ for the effective hyperplanes and use the global convention

$$s(M) = \begin{cases} \max_{H \in \mathcal{H}(M)} |H|, & \mathcal{H}(M) \neq \emptyset, \\ r-1, & \mathcal{H}(M) = \emptyset, \end{cases} \quad D_u = n - \left\lceil \frac{n}{r} \right\rceil. \quad (1.0)$$

Theorem 1.1 (fixed rank).

For every fixed integer $r \geq 3$, there is an integer N_r such that, for every $n \geq N_r$ and every n -element rank- r paving matroid M , all zeros of $h_M^*(x)$ are simple, strictly negative, and real.

The threshold may depend on r ; it is not made explicit. The theorem does not include rank two or assert anything for all small n .

Theorem 1.2 (growing rank).

Let $n_\nu \rightarrow \infty$ and $r_\nu \rightarrow \infty$. Put

$$D_\nu = n_\nu - \left\lfloor \frac{n_\nu}{r_\nu} \right\rfloor, \quad X_\nu = \frac{n_\nu}{r_\nu^4}, \quad L_\nu = \log X_\nu, \quad A_\nu = \log(2r_\nu + L_\nu).$$

Assume

$$\frac{r_\nu^4 A_\nu^4}{L_\nu} \rightarrow 0. \quad (1.1)$$

Let M_ν be a connected n_ν -element rank- r_ν paving matroid, and suppose every effective hyperplane of M_ν has size at most $D_\nu - 1$. Then, for all sufficiently large ν ,

$$\deg h_{M_\nu}^* = D_\nu$$

and $h_{M_\nu}^*$ has at least

$$D_\nu - J_\nu, \quad J_\nu = \lceil r_\nu^7 A_\nu^4 \rceil, \quad (1.2)$$

pairwise distinct strictly negative real zeros. Moreover $J_\nu = o(D_\nu)$.

We do not assert that the zeros counted in Theorem 1.2 are simple, or that the remaining J_ν zeros are real, negative, or distinct.

2 The exact paving correction formula and the uniform phase grid

All analytic formulas in Sections 2–6 are first written for a connected matroid, so that $\dim P(M) = n - 1$ and the intrinsic Ehrhart denominator is $(1 - x)^n$. Disconnected paving matroids are classified and reduced by an integral affine isomorphism at the end of the fixed-rank proof.

Put $A = 1 - x$, $m = n - 1$, and, for every polynomial $f(k)$, write

$$\mathcal{E}_n(f) = A^n \sum_{k \geq 0} f(k) x^k, \quad U_{\ell,n}(x) = \mathcal{E}_n \binom{\ell k + n - 1}{n - 1}. \quad (2.1)$$

Here $U_{r,n}(x)$ denotes only the leading Veronese term, not the full uniform-matroid numerator $h_{U_{r,n}}^*(x)$. The latter also contains the lower-Euler terms in the first sum of (2.5). To define the hyperplane correction, let $M_{\ell,0}(x)$ be the $\ell \times \ell$ matrix, indexed by $0, \dots, \ell - 1$, whose (i, j) -entry is 1 if $i \geq j$ and x if $i < j$. Choose any nonzero index a_0 , let $D_{a_0}(u)$ have entry 1 at a_0 and u at every other diagonal position, and set

$$B_m^{(\ell)}(u, x) = e_0^\top (M_{\ell,0}(x) D_{a_0}(u))^m \mathbf{1}, \quad K_{\ell,m,s}(x) = \sum_{a=0}^{s-1} [u^a] B_m^{(\ell)}(u, x). \quad (2.2)$$

We abbreviate $M_\ell(u, x) = M_{\ell,0}(x) D_{a_0}(u)$, and write $M(u)$ when ℓ, x are fixed. The choice of $a_0 \neq 0$ does not change the coefficient identity used here. Indeed,

$$[u^a] B_m^{(\ell)}(u, x) = A^{m+1} \sum_{k \geq 0} \binom{(\ell - 1)k + a}{a} \binom{k + m - a - 1}{m - a} x^k. \quad (2.3)$$

For an effective hyperplane H of size s , put $q = n - s$. For $0 \leq h \leq r - 1$, let $\ell = r - h$ and

$$\begin{aligned} W_h(k) &= \binom{\ell k + m - h}{m}, \\ Q_{h,H}(k) &= \sum_{J=0}^{k-1} \binom{J+q-1}{q-1} \binom{\ell k - h - J + s - 1}{s-1}. \end{aligned} \quad (2.4)$$

Hanely et al., Proposition 5.10 and Theorem 5.11, give the underlying Ehrhart polynomial formula. Multiplying its lattice-point series by $(1-x)^n$, which is the intrinsic denominator in the connected case, gives the following full coordinate inclusion–exclusion form:

$$h_M^*(x) = \sum_{h=0}^{r-1} (-1)^h \binom{n}{h} \mathcal{E}_n(W_h) - \sum_H \sum_{h=0}^{r-1} (-1)^h \binom{s_H}{h} \mathcal{E}_n(Q_{h,H}). \quad (2.5)$$

For $0 \leq h \leq r - 2$, the exact finite-band identities are

$$\begin{aligned} \mathcal{E}_n(Q_{h,H}) &= \sum_{j=0}^h (-1)^j \binom{h}{j} A^j K_{r-h,m-j,s_H-j}, \\ \mathcal{E}_n(W_h) &= \sum_{j=0}^h (-1)^j \binom{h}{j} A^j U_{r-h,n-j}. \end{aligned} \quad (2.6)$$

When $h = r - 1$, one has $Q_{h,H} = W_h$ and $\mathcal{E}_n(W_h) = x^{r-1}$; this one-color term is kept separately. The $h = 0$ terms in (2.5) give the exact decomposition

$$h_M^*(x) = U_{r,n}(x) - \sum_H K_{r,n-1,s_H}(x) + L_M(x), \quad (2.7)$$

where L_M is now defined, rather than suppressed, by the terms with $h \geq 1$ in (2.5)–(2.6). Distinct bad caps are disjoint in the coordinate-bounded domain because two simultaneous cap violations would force the total coordinate sum to exceed rk ; this is the step that makes the sum over H in (2.5) exact.

We use two packing consequences repeatedly:

$$\sum_H \binom{s_H}{r-1} \leq \binom{n}{r-1}, \quad (2.8)$$

and, for every fixed r ,

$$\sum_H s_H^{r-1} \leq C_r n^{r-1}. \quad (2.9)$$

In particular, for each fixed $\delta > 0$, the number of hyperplanes with $s_H > \delta n$ is bounded in terms of r, δ , while in the growing-rank argument it is at most $(C/\delta)^r$.

At $x = -t^r$, the unperturbed matrix in (2.2) is similar to a normal circulant. Its eigenvalues are

$$\lambda_\omega(t) = \frac{1+t^r}{1-te^{\pi i/r}\omega}, \quad \omega^r = 1. \quad (2.10)$$

Exactly two conjugate branches, denoted τ_+ and τ_- , have maximal modulus. We write S_3 for the third singular value of the transfer matrix and set

$$d_r(t) = |1 - te^{-\pi i/r}|, \quad R_r(t) = \frac{1+t^r}{d_r(t)}, \quad 1 - te^{-\pi i/r} = d_r(t)e^{i\phi_r(t)}, \quad 0 < \phi_r(t) < \frac{r-1}{r}\pi. \quad (2.11)$$

The two dominant Fourier branches of the leading section $U_{r,n}$ have equal modulus $R_r(t)^n$. Their phases lead to the grid

$$n\phi_r(t_j) = j\pi. \quad (2.12)$$

At a grid point, the two principal branches contribute $2(-1)^j R_r(t_j)^n / r$. All sign arguments below consist in showing that the total hyperplane correction and all lower-Euler branches cannot erase this alternating main term.

The phase has the endpoint expansions

$$\phi_r(t) = t \sin(\pi/r) + O_r(t^2) \quad (t \downarrow 0), \quad (2.13)$$

and, with $p = (r-1)/r$,

$$p\pi - \phi_r(t) = \frac{\sin(\pi/r)}{t} + \frac{\sin(\pi/r) \cos(\pi/r)}{t^2} + O_r(t^{-3}) \quad (t \rightarrow \infty). \quad (2.14)$$

The uniform two-branch phase-grid argument follows the hypersimplex prototype in Adiprasito–Zhang [1, Sections 3.4–3.5]. The estimates below control the nonuniform paving corrections; the uniform phase mechanism itself is not claimed as new.

3 Quantitative sign control

This section packages the analytic estimates used in the growing-rank proof. They are also the local ingredients of the fixed-rank argument.

Proposition 3.0 (the ultra-small direct endpoint).

Let $r, n \rightarrow \infty$, put $m = n-1$, and let $t_j \leq 1/4$ be a phase-grid point. Set $\kappa = mt_j$. Suppose, with fixed absolute constants C_*, C_1 , that

$$\begin{aligned} \frac{\kappa}{r^8 \log^2(2r)} &\rightarrow \infty, & \frac{\kappa}{r^6 \log^6(1+\kappa)} &\rightarrow \infty, \\ r^2 + r \log(2r) &= o(\log(1/t_j)), \\ C_*^r t_j \log(1+\kappa) &= o(1/r), & (C_1 r)^r C_*^r t_j \log(1+\kappa) &= o(1/r). \end{aligned} \quad (3.0)$$

If every effective hyperplane has size at most $D-1 = n - \lceil n/r \rceil - 1$, then

$$(-1)^j \frac{h_M^*(-t_j^r)}{R_r(t_j)^n} \geq \frac{1}{2r}. \quad (3.0a)$$

Proof. Take $\delta = 1/[1000(r+1)]$. The paving packing inequality gives at most $(C_1 r)^r$ hyperplanes with $s_H > \delta n$, while $\sum_H (s_H)_{r-1} \leq 4^{r-1} n^{r-1}$. The small hyperplanes and every lower-Euler layer in (2.5)–(2.6) have total normalized size

$$\exp\{Cr^2 \log(2r)\} (1+\kappa)^{2r} e^{-c\kappa} = o(1/r). \quad (3.0b)$$

All but at most one large hyperplane are separated from the critical density. For each such term, the central tail, middle arc, and far arc give

$$C_*^r t_j \log(1+\kappa) + P_r(\kappa) e^{-c\sqrt{\kappa}} + P_r(\kappa) e^{-c\kappa/r^2 + C\sqrt{\kappa}} + P_r(\kappa) e^{-c\kappa/(r^4 \log(2r)) + C\sqrt{\kappa}}, \quad (3.0c)$$

where $\deg P_r \leq Cr$ and its coefficient height is $\exp(O(r^2 \log(2r)))$. The five assumptions in (3.0), including the last one after multiplication by $(C_1 r)^r$, make their full sum $o(1/r)$. For the unique possible critical hyperplane, the complex Gaussian cumulative has modulus at most $1/2$, so its signed contribution is at most $1/r + o(1/r)$. The uniform pair contributes $2/r$, proving (3.0a). $\square$

Proposition 3.1 (central and noncentral hyperplane estimates).

Let $r \geq 4$, $m = n - 1$, and let $s \leq (r - 1)m/r$. Put

$$\xi = \frac{s - (r - 1)m/r}{\sqrt{m}}.$$

On the phase grid (2.12), the normalized highest-Euler correction has the central approximation

$$\frac{K_{r,m,s}(-t^r)}{R_r(t)^n} = \frac{2}{r} \operatorname{Re} \left[e^{in\phi_r(t)} \Phi \left(\frac{\xi}{\sqrt{v_r(t)}} \right) \right] + E_{r,n,t,s}, \quad (3.1)$$

where

$$v_r(t) = \frac{(r - 1)/r}{rt e^{\pi i/r}}, \quad (3.2)$$

and the branch of the Gaussian distribution function satisfies

$$\left| \Phi \left(\frac{\xi}{\sqrt{v_r(t)}} \right) \right| \leq \frac{1}{2}. \quad (3.3)$$

The error in (3.1) is $o(1/r)$, uniformly in the following three regions, under the corresponding assumptions:

1. $t_- \leq t \leq 1/4$, where $t_- = n^{-1/(K_0 r)}$ and $r^3 \log(2r) = o(\log n)$;

2. $1/4 \leq t \leq 4$, where

$$e^{C_0 r \log(2r)} r^{12} \log^3(n + 3) = o(n); \quad (3.4)$$

3. $t \geq 4$, with $\kappa = (n - 1)/t$, where

$$\frac{\kappa}{r^6 \log^3(2r + \kappa)} \rightarrow \infty \quad (3.5)$$

and (3.4) holds.

For hyperplanes separated from the critical ratio $(r - 1)/r$, the right side is $o(1/r)$ after summation over all such hyperplanes. At most one effective hyperplane can lie in the critical window.

Proof. The content transfer matrix at $x = -t^r$ has two maximal-modulus Fourier eigenvalues and $r - 2$ subordinate eigenvalues. On $1/4 \leq t \leq 4$, the third singular value S_3 and the two dominant phases satisfy

$$1 - (S_3/R_r)^2 \geq c/r^2, \quad |\tau_+ - \tau_-| \geq c/r. \quad (3.6)$$

If $u = e^{i\theta}$ is the coefficient-extraction variable, the rank-one diagonal perturbation and projection onto the dominant two-dimensional Fourier space give

$$\frac{\operatorname{spr} M(u)}{R_r} \leq 1 - c \min\{\theta^2/r^3, r^{-4}\}. \quad (3.7)$$

Schur triangularization then yields

$$\|(M(u)/R_r)^m\| \leq \exp\{Cr \log(m + r) - cm \min(\theta^2/r^3, r^{-4})\}. \quad (3.8)$$

In a c/r^2 neighborhood of $u = 1$, disjoint root circles and the resolvent formula bound the first three logarithmic derivatives of the two principal branches and their residual amplitudes by

$$Q_r \leq (Cr)^{Cr}. \quad (3.9)$$

Using the inward contour $u = \exp((-h + iy)/\sqrt{m})$, completing the quadratic exponent, and using (3.8) off the central arc gives (3.1). The argument of $-\xi/\sqrt{2v_r(t)}$ lies in the positive-kernel sector, which proves (3.3).

For $0 < t \leq 1/4$, the corresponding estimates are

$$1 - (S_3/R_r)^2 \geq ct/r^2, \quad |\tau_+ - \tau_-| \geq ct/r, \quad (3.10)$$

and hence

$$\text{spr}(M(e^{i\theta}))/R_r \leq 1 - c \min\{t\theta^2/r^3, t^3/r^4\}. \quad (3.11)$$

The analytic envelope is at most $(Cr/t)^{Cr}$. Taking $t \geq n^{-1/(K_0 r)}$ with K_0 sufficiently large makes the central Taylor error and the outer-arc loss uniform, proving the first region.

For $t \geq 4$, scale the characteristic polynomial by $\lambda = t^{r-1}\mu$, $\varepsilon = 1/t$. It becomes

$$Q_{r,t}(\mu, u) = (\varepsilon\mu - a_0)(\varepsilon\mu - ua_0)^{r-1} + \mu^r, \quad a_0 = 1 + t^{-r}. \quad (3.12)$$

Its limiting roots satisfy $\nu_j^r = (-1)^{r+1}u^{r-1}$. Rouché circles of radius c/r and one step of perturbation theory give

$$\mu_j = \nu_j - \frac{1}{t} \frac{\nu_j^2}{r} \left(1 + \frac{r-1}{u}\right) + O(Q_r/t^2), \quad (3.13)$$

with the same bound after three logarithmic derivatives. Consequently the principal modulus loss is at least $c\theta^2/(rt)$ near zero and $c/(tr^3)$ away from zero. The contour scale $r\sqrt{B \log(2r + \kappa)}/\sqrt{\kappa}$, for a fixed profile constant B , and (3.5) now give (3.1).

Finally, (2.8) implies that hyperplanes above a fixed density have cardinality at most $(Cr)^r$. Two hyperplanes cannot both meet the critical density window, because their intersection would exceed $r - 2$. The retained Gaussian left-tail factor for every noncritical term absorbs this count. Small hyperplanes are summed before normalization, using (2.8), rather than by multiplying a pointwise $o(1)$ estimate. This proves the proposition. $\square$

Proposition 3.2 (low-Euler suppression).

Under the hypotheses of each of the three regions in Proposition 3.1,

$$\frac{L_M(-t^r)}{R_r(t)^n} = o(1/r) \quad (3.14)$$

uniformly over the indicated paving matroids.

Proof. The layer of Euler index $r - h$ has spectral-radius ratio

$$\frac{R_{r-h}}{R_r} \leq \begin{cases} \exp(-ct/r^3), & 0 < t \leq 1/4, \\ \exp(-c/r^3), & 1/4 \leq t \leq 4, \\ \exp(-c/(tr^3)), & 4 \leq t \leq T_r, \\ \exp(-c \log t/r), & t \geq T_r, \end{cases} \quad (3.15)$$

for an exponential threshold T_r . The finite-band, inclusion-exclusion, and hyperplane-packing factors contribute at most

$$\exp\{Cr \log n + Cr^2 \log(2r) + Cr^2 \log^+ t\}. \quad (3.16)$$

The assumptions in the respective regions make the n -fold loss from (3.15) dominate (3.16). The isolated one-color term has the stronger bound $\exp(Cr \log n + Cr^2 \log t - cnr \log t)$ in the large- t region. Thus all lower-Euler layers sum to (3.14). $\square$

Corollary 3.3 (alternating grid signs).

Whenever the hypotheses of Propositions 3.1 and 3.2 hold at a phase-grid point t_j ,

$$(-1)^j \frac{h_M^*(-t_j^r)}{R_r(t_j)^n} \geq \frac{1}{2r} \quad (3.17)$$

for all sufficiently large parameters.

Proof. The uniform pair contributes $2/r$. The unique possible critical hyperplane contributes at most $1/r + o(1/r)$, while all noncritical hyperplanes and all lower-Euler layers contribute $o(1/r)$. $\square$

Proposition 3.3A (bounded critical-excess middle signs).

Fix $r \geq 4$ and $C \geq 0$. Put

$$D_u = n - \left\lceil \frac{n}{r} \right\rceil, \quad s = D_u + \delta, \quad 0 \leq \delta \leq C. \quad (3.17a)$$

For all sufficiently large n , every connected rank- r paving matroid with largest effective hyperplane of size s has the alternating sign (3.17) at every phase-grid point between the fixed direct and reciprocal endpoint circles. The threshold is uniform in the matroid and in $\delta \in \{0, \dots, C\}$.

Proof. Write

$$\begin{aligned} b_u &= r \lceil n/r \rceil - n, \\ \ell &= \left\lceil \frac{\delta + 1 - b_u}{r - 1} \right\rceil, & a &= (r - 1)\ell + b_u - \delta - 1. \\ c &= \lceil n/r \rceil + \ell, & b &= b_u + r\ell, \end{aligned} \quad (3.17b)$$

Thus $a \in \{0, \dots, r-2\}$, while a, b, ℓ range over a finite set depending only on r, C . Insert the shifted highest cutoff into the exact finite-band formula (2.5)–(2.7). Relative to the central cutoff in Proposition 3.1, the two-moving local expansion is multiplied by $\exp(O(\sigma)) = 1 + o(1)$, uniformly because the shift is bounded. The central Gaussian parameter still satisfies $\xi \rightarrow 0$. The packing estimate for every additional hyperplane and the complete lower-Euler estimate are unchanged, and hence their sum is $o(1/r)$. The uniform pair therefore contributes $2/r + o(1/r)$, the unique possible critical hyperplane at most $1/r + o(1/r)$, and all remaining terms $o(1/r)$. This yields the strict margin $1/(2r)$ on the whole middle grid.

This is an independent bounded-increment estimate, not an application of Proposition 3.1 outside its stated range. The complete Taylor bounds, including the two-moving remainder and lower-Euler sum, are in proof-dossier D2 under S59 and S91. Since only finitely many (a, b, ℓ) occur, their thresholds have a common maximum. $\square$

Proposition 3.4 (the giant-profile middle sign).

Fix $r \geq 5$, put $d = r - 1$, and suppose the largest effective hyperplane is H_0 of size $n - q$, where $2 \leq q < n/r$. Set $p = q/n$, $a_0 = 1 - rp$, and let $W = he^{i\theta}$, $\pi/r < \theta < \pi/d$, be the leading solution of

$$(1 - p)W^r + a_0 t^r W + dp t^r = 0. \quad (3.18)$$

In the separated subcritical range $p \leq p_{\max} < 1/r$, define

$$\chi_n(t) = n[-\arg(1 - W) + p \arg(dp + a_0 W)] - \frac{\theta}{2} - \frac{1}{2} \arg(rp + a_0 W). \quad (3.19)$$

There is a fixed-rank threshold $L = L(r, p_{\max})$ such that, whenever

$$q, \quad n - rq, \quad nh, \quad n/h \geq L, \quad (3.20)$$

every point satisfying $\chi_n(t) = \ell\pi$ has $\text{sign } h_M^*(-t^r) = (-1)^\ell$. The statement includes every legal collection of additional hyperplanes.

In the moving critical range $p \rightarrow 1/r$, $n - rq \rightarrow \infty$, put

$$U_n(t) = n[-\arg(1 - W) + p \arg(dp + a_0 W)]. \quad (3.21)$$

There is a threshold L_r such that the same four inequalities (3.20) and $U_n(t) = \ell\pi$ imply the same sign. No hypothesis that $n(1 - rp)^2 \rightarrow \infty$ is required.

Proof. Put

$$K = \frac{a_0 h}{dp}, \quad V = p(1 - p) \left(a_0 + \frac{rp}{W} \right), \quad \Sigma = K \sqrt{n \operatorname{Re} V}. \quad (3.21a)$$

The exact contour obtained from (2.5)–(2.7) is split into the two leading saddle basins, their bounded connecting arc, and the remaining arcs. The full estimates in proof-dossier D4 give, uniformly over every legal family of additional hyperplanes,

$$h_M^*(-t^r) = 2\mathcal{M} \cos \chi_n + E, \quad \frac{|E|}{\mathcal{M}} \leq \varepsilon_r(q, n - rq, nh, n/h, \Sigma), \quad (3.21b)$$

where $\mathcal{M} > 0$, and $\varepsilon_r \rightarrow 0$ whenever its five displayed arguments tend to infinity. This is a full-contour estimate: the central arc uses the quadratic spectral gap, the bounded connecting arc uses the exact cumulative integral, and the far arcs use the strict full-circle minimum-modulus gap. Packing is applied before summing the additional-hyperplane corrections, so no factor equal to their number remains. In the subcritical range, $\Sigma^2 \geq c_{r,p_{\max}} nh$; hence the four inequalities (3.20) make the right side of (3.21b) smaller than the fixed positive grid margin. This proves the first clause.

For the moving critical range, a sequence violating the statement has a subsequence on which Σ either diverges or stays bounded. The first case again follows from (3.21b). In the bounded case, the exact moving-saddle relations imply $a_0 \rightarrow 0$, $h/t \rightarrow 1$, $na_0^2 t = O(1)$, and $a_0(1 + t) \rightarrow 0$. Taylor expansion of the Legendre exponent, uniformly on the real parameter segment, gives

$$n\{S(p) - S(1/r)\} = -\frac{n(1 - rp)^2 W_0}{2d} + O(n(1 - rp)^3 t(1 + t)), \quad W_0 = te^{\pi i/r}. \quad (3.22)$$

With

$$z = -\frac{\xi}{\sqrt{2v_r(t)}}, \quad \xi = -\frac{na_0 + d}{r} \sqrt{\frac{t}{n-1}}, \quad (3.22a)$$

this is equivalent to

$$U_n(t) = n\phi_r(t) - \operatorname{Im} z^2 + o(1). \quad (3.22b)$$

The remaining scaled complementary-error factor is $E(z) = e^{z^2} \operatorname{erfc}(z)$, with $\arg z = \pi/(2r)$. Its exact integral representation

$$E(z) = \frac{2}{\pi} \int_0^\infty \frac{ze^{-u^2}}{u^2 + z^2} du \quad (3.22c)$$

puts every integrand in the sector $[-\pi/(2r), \pi/(2r)]$. Thus $\operatorname{Re} E(z) \geq \cos(\pi/(2r))|E(z)| > 0$, uniformly on each bounded separation window. At $U_n = \ell\pi$, (3.22b)–(3.22c) therefore give $\operatorname{sign}(-1)^\ell$. Divergent and bounded Σ exhaust every failure sequence, so the sequential estimate is equivalent to one finite threshold L_r for the four quantities in (3.20). $\square$

Proposition 3.5 (rank-four giant-profile middle signs).

Let $r = 4$, $d = 3$, and let the largest effective hyperplane have size $n - q$. Define $p = q/n$, $a_0 = 1 - 4p$, the leading solution $W = he^{i\theta}$ by (3.18), and the phases χ_n and U_n by (3.19) and (3.21). There are finite gates with the following properties.

1. For each $p_{\max} < 1/4$, if q , nh , and n/h exceed the gate, then every $\chi_n(t) = \ell\pi$ between the endpoint circles has sign $(-1)^\ell$.
2. If q , $n - 4q$, nh , and n/h exceed the critical gate, then every $U_n(t) = \ell\pi$ between the endpoint circles has sign $(-1)^\ell$.

Consequently these signs hold for every actual connected sequence in one of the three ranges

$$q \rightarrow \infty, \quad q/n \rightarrow 0, \quad q/n \rightarrow p_0 \in (0, 1/4), \quad q/n \rightarrow 1/4 \text{ with } n - 4q \rightarrow \infty, \quad (3.23)$$

uniformly over all additional hyperplanes.

Proof. For the separated regime, the exact four-state transfer matrix has a simple leading conjugate pair on the full contour. After normalizing by its leading modulus, the other two eigenvalues and every attached-hyperplane correction satisfy

$$\frac{|E|}{\mathcal{M}} \leq \varepsilon_4(q, n - 4q, nh, n/h), \quad (3.24)$$

where the right side tends to zero when the displayed active arguments do. This estimate is obtained by the same basin/connecting-arc/far-arc partition as (3.21b), but with the quartic characteristic polynomial retained exactly; the full-circle spectral gap and the resolvent bounds are in proof-dossier D5, S50–S54. At a χ_n -grid point the normalized leading pair is $2 \cos \chi_n$, so (3.24) proves the first clause.

When $p \rightarrow 1/4$, write again $z = -\xi/\sqrt{2v_4(t)}$. The exact rank-four Legendre calculation gives

$$U_n(t) = n\phi_4(t) - \operatorname{Im} z^2 + o(1), \quad (3.25)$$

uniformly in the bounded separation window; the remainder is controlled by the four quantities in the second clause, without an additional Σ gate. The scaled factor is $e^{z^2} \operatorname{erfc} z$, and the sector argument (3.22c) gives its strictly positive real part. This proves the critical clause. If $q/n \rightarrow 0$ with $q \rightarrow \infty$, use the separated clause after the sparse-outside rescaling; bounded q is handled separately by Proposition 4.4. The paving intersection bound is imposed before summing the exact corrections, so (3.24)–(3.25) concern the complete polynomial. □

4 Endpoint root blocks

The sign grid does not by itself reach either endpoint. We use reciprocal scalings and Rouché transfer.

Proposition 4.0 (fixed-rank direct endpoint).

Fix $r \geq 4$, put $p = (r - 1)/r$, and let $\alpha = (\alpha_1, \alpha_2, \dots)$ be a finite or countable sequence with $\alpha_i \geq 0$, $\sum_i \alpha_i \leq 1$, and $\max_i \alpha_i \leq p$. If $(N_1, N_2, \dots, N_0)$ has multinomial distribution with rj trials and cell probabilities $(\alpha_1, \alpha_2, \dots, 1 - \sum_i \alpha_i)$, define

$$F_\alpha(z) = \sum_{j \geq 0} \Pr\{N_i \leq (r - 1)j \text{ for every } i\} \frac{z^j}{(rj)!}. \quad (4.0)$$

Every F_α has infinitely many simple strictly negative zeros. For every sufficiently large fixed j , the circle

$$|z| = T_j^r, \quad T_j = \frac{\pi j}{\sin(\pi/r)}, \quad (4.0a)$$

contains exactly j zeros and has no boundary zero, uniformly over all such spectra. If M_n is any connected fixed-rank paving sequence whose effective-hyperplane densities converge to α , then $h_{M_n}^*(z/n^r) \rightarrow F_\alpha(z)$ locally uniformly. Hence, uniformly over the actual objects in Lemma 5.1, the interval $0 < t < t_{n,j}$, where $n\phi_r(t_{n,j}) = j\pi$, contains exactly j simple negative roots and the boundary value has sign $(-1)^j$.

Proof. Formula (2.5) gives the coefficient probability in (4.0), and factorial domination gives compact convergence, including countable spectra. We first prove the root statement without using the circles. Put $d = r - 1$, order the positive α_i , and set

$$h(y) = \frac{y}{ry - d}, \quad f(t) = \frac{t}{d - rt}, \quad N(t) = \#\{i : \alpha_i > t\}. \quad (4.0b)$$

Root-of-unity filtering and the common two-branch change of variables transform the normalized coefficient series into a cosine transform with signed weight H . Its downward variation is exactly

$$V_- = \sum_{i \geq 2} f(\alpha_i) - h(p_b) + 1, \quad b = \alpha_2, \quad (4.0c)$$

with the empty or one-atom cases obtained by deletion. Here $p_b \in (d/r, 1)$ is the unique high-branch point paired with the low-branch point b , namely $b^d(1 - b) = p_b^d(1 - p_b)$. Thus $h(p_b) > 1$ for $b > 0$. Since $\sum_{i \geq 2} \alpha_i \leq 1 - b$ and $\alpha_i \leq b \leq 1/2$,

$$V_- < \frac{1 - b}{d - rb} \leq \frac{1}{r - 2} \leq \frac{1}{2}. \quad (4.0d)$$

All jumps of H are upward, $H(0) = 1$, and the possible terminal inverse-square-root singularity is integrable. Truncating that singularity does not increase V_- . Hence the signed-cosine rectangle argument has strict vertical and horizontal margins and puts the normalized entire function C in the Laguerre–Pólya class. The exact gamma multiplier

$$\gamma_j = \frac{(2j)!}{(dj)!j!} \quad (4.0e)$$

takes C to F_α ; its multiplier-sequence strictification makes every zero simple and negative. The coefficient is strictly positive, the order is at most $1/r < 1$, and the function is not a polynomial, so there are infinitely many zeros. The countable case follows by truncation: $tN(t) \rightarrow 0$, the total jump mass is finite, and the kernel integrals are dominated by $\sum_i \alpha_i^{1+d/2} < \infty$.

Now all zeros are already known to be simple and negative. The explicit root-of-unity formula for the unperturbed kernel gives the sign $(-1)^j$ at $-T_j^r$. The same rectangle margins and (4.0d) make that sign uniform over the compact spectrum space for all sufficiently large fixed j . Thus the circle $|z| = T_j^r$ is zero-free: any boundary zero would have to be the negative point already excluded by the strict sign. The negative-axis sign changes provide j real zeros inside, and the argument principle gives the same count. Local uniform convergence of actual rescaled polynomials and small disjoint Rouché disks transfer the count and simplicity. A failure sequence has a convergent spectrum subsequence and contradicts this zero-free circle. The full cosine-transform calculation and the uniform rectangle estimates are proof-dossier D4. $\square$

Proposition 4.1 (fixed-rank endpoint transfer).

Fix $r \geq 4$, let $b = r\lceil n/r \rceil - n$, and suppose first that the largest effective hyperplane has size $s \leq D_u - 1$, where $D_u = n - \lceil n/r \rceil$. Put $g = D_u - s \geq 1$. The reciprocal endpoint kernel is

$$F_{r,b,g}(z) = \sum_{j \geq 0} \Pr \left\{ \text{Bin} \left(rj + b, \frac{r-1}{r} \right) \leq (r-1)j + b + g - 1 \right\} \frac{z^j}{(rj + b)!}. \quad (4.1)$$

For every $b \in \{0, \dots, r-1\}$ and $g \geq 1$, this entire function has infinitely many simple strictly negative zeros. As $g \rightarrow \infty$, it converges locally uniformly to

$$V_{r,b}(z) = \sum_{j \geq 0} \frac{z^j}{(rj + b)!}, \quad (4.2)$$

which has the same zero property. This limiting function is the residue function E_b of Adiprasito–Zhang [1, Section 3.3], with $k = r$; its simple negative zeros are prior work. Here the additional assertions concern the corrected kernels $F_{r,b,g}$ and counts uniform in g . For every sufficiently large fixed k , the circles

$$|z| = T_{k,b}^r, \quad T_{k,b} = \frac{\pi(k + b/r)}{\sin(\pi/r)}, \quad (4.3)$$

contain exactly k zeros, have no boundary zero, and work uniformly in g , including the limit $g = \infty$.

If instead $0 \leq s - D_u \leq C$, the actual codegree differs from $\lceil n/r \rceil$ by a bounded integer. For fixed r, C , the residue b and the additional endpoint index a range over a finite set. Denote by $E_{r;b,a}$ the entire function obtained as the locally uniform limit of the exact reciprocal polynomial (2.5) in that fixed (b, a) -subsequence. Every such $E_{r;b,a}$ has infinitely many simple strictly negative zeros and zero-free counting circles of the form (4.3). Consequently every fixed finite block of direct and reciprocal endpoint zeros transfers uniformly to simple negative zeros of h_M^* .

Proof. Coefficientwise convergence follows from (2.3)–(2.7), and uniform factorial tails make it locally uniform at the origin. The sign of $F_{r,b,g}(-T_{k,b}^r)$ is $(-1)^k$. A full-circle argument-principle count gives exactly k zeros inside (4.3); hence the alternating real zeros already supplied by the signs exhaust the disk and are simple. The estimates are uniform in g : if a common transfer threshold failed, a failure sequence would have, after fixing the finite residue b , either a constant subsequence $g = g_0$ or a subsequence $g \rightarrow \infty$. The corresponding locally uniform limit is respectively F_{r,b,g_0} or $V_{r,b}$, and the same zero-free circle contradicts failure by Rouché’s theorem. Thus no maximum over infinitely many individual g -thresholds is taken. The bounded-excess kernels form a genuinely finite (b, a) -family, so the same argument takes a finite maximum there. $\square$

Proposition 4.2 (growing-rank reciprocal block).

Under (1.1), let

$$K = \lceil r^5 A^4 \rceil, \quad K_{\max} = \left\lfloor \frac{r}{16} L \right\rfloor. \quad (4.4)$$

Set $b = r \lceil n/r \rceil - n$ and define

$$Q_n(z) = n^{-b} \left(\frac{z}{n^r} \right)^D h_M^*(n^r/z). \quad (4.5)$$

Then $K/K_{\max} \rightarrow 0$, and Q_n has exactly $K - 1$ simple strictly negative zeros in $|z| < T_{K-1,b}^r$, with no zero on the boundary and boundary sign $(-1)^{K-1}$. Under (4.5), these correspond to $K - 1$ zeros of h_M^* in $x < -n^r/T_{K-1,b}^r$.

Proof. For the first $2K$ reciprocal coefficients, the paving correction formula has the same leading factorial kernel as the uniform polynomial, with a relative error tending to zero uniformly in the hyperplane family. On the circle of radius $T_{K-1,b}^r$, with $T_{k,b}$ as in (4.3), the coefficient tail is split at the central index. Below it, the factorial ratio is geometric after the K -scale; above it, the reciprocal factorial tail is dominated by its first omitted term. Condition (1.1) implies $K = o(rL)$, so the two estimates overlap. More precisely, if

$$G_{r,b,K}(z) = \sum_{j=0}^{2K} \frac{z^j}{(rj + b)!}, \quad \Gamma_K = \{|z| = T_{K-1,b}^r\}, \quad (4.5a)$$

then the two coefficient ranges and their tails give

$$\sup_{z \in \Gamma_K} |Q_n(z) - G_{r,b,K}(z)| \leq \eta_n \inf_{z \in \Gamma_K} |G_{r,b,K}(z)|, \quad \eta_n \rightarrow 0. \quad (4.5b)$$

The denominator in (4.5b) is positive by the full root-of-unity circle estimate, not by a negative-axis test. The coefficient-ratio derivation of (4.5b), including the uniform paving correction and both factorial tails, is proof-dossier D6. Rouché's theorem now gives the exact root count and boundary sign for Q_n . Multiplication by the monomial used in (4.5) changes the boundary sign by the known parity $(-1)^D$, which is the conversion used in (6.7). $\square$

Proposition 4.3 (fixed-rank giant-profile endpoint blocks).

Fix $r \geq 4$, put $d = r - 1$, and suppose the largest effective hyperplane has size $n - q$, where $2 \leq q < n/r$. With $p = q/n$, $a_0 = 1 - rp$, and the leading solution $W = he^{i\theta}$ of (3.18), put

$$c = \left\lceil \frac{n - q + 1}{r - 1} \right\rceil, \quad b = rc - n, \quad a = (r - 1)c - (n - q) - 1, \quad D = n - c. \quad (4.6)$$

Then $a \in \{0, \dots, r - 2\}$, the codegree is exactly c , and the degree is exactly D . In the subcritical range $q \rightarrow \infty$, $q/n \leq p_{\max} < 1/r$, every sufficiently large fixed j, k admits disjoint boundaries

$$x_L = -\frac{Y_j}{p(1-p)^{r-1}n^r}, \quad x_U = -\frac{\Lambda}{T_{a,k}^{r-1}}, \quad \Lambda = \frac{(n-q)^{r-1}(q+b)}{b}, \quad (4.7)$$

such that the disk ending at x_L contains exactly j simple negative roots and has sign $(-1)^j$, while the exterior beyond x_U contains exactly k simple negative roots and has sign $(-1)^{D-k}$. Here Y_j is the fixed Wright-phase radius with its phase equal to $j\pi$, and $T_{a,k} = \pi(k + a/(r-1))/\sin(\pi/(r-1))$.

In the critical range $q/n \rightarrow 1/r$, $n - rq \rightarrow \infty$, the same conclusion holds with the direct boundary $x_L = -R_{p,j}/n^r$, where the fixed macroscopic circle $R_{p,j}$ contains exactly j roots, and with the same reciprocal boundary (4.7). At these boundaries,

$$U_n(t_L) = j\pi + o(1), \quad U_n(t_U) = (D - k)\pi + \frac{\pi}{r-1} + o(1). \quad (4.8)$$

Proof. In the subcritical range, multiplication of the exact saddle equation by n^r at the scale $t^r = Y/[p(1-p)^{r-1}n^r]$ gives

$$nW \longrightarrow \frac{d^{1/r}Y^{1/r}}{1-p_0} e^{\pi i/r}. \quad (4.8a)$$

The exact coefficient formula and factorial domination therefore give local uniform convergence of the direct polynomial to the Wright kernel whose continuous phase Ψ satisfies $\Psi(Y_j) = j\pi$. Its full Hankel-circle estimate has one strictly dominant conjugate pair on the boundary and hence exactly j simple negative zeros inside. The remaining macroscopic hyperplanes contribute an integrable signed weight whose downward variation is strictly below one half; thus the same boundary is zero-free for the complete kernel. The convergence is uniform in all legal spectra, so Rouché transfers the count to the actual polynomial.

At the reciprocal scale, after fixing $a \in \{0, \dots, d-1\}$, the normalized polynomial converges locally uniformly to

$$V_{d,a}(z) = \sum_{m \geq 0} \frac{z^m}{(dm + a)!}. \quad (4.8b)$$

The function (4.8b) is likewise the residue function of [1, Section 3.3], with $k = d$. We use its known simple negative zeros as the uniform endpoint baseline, rather than claim a new zero theorem for that function. The root-of-unity filter shows that the circle $|z| = T_{a,k}^d$ has one dominant conjugate pair and is zero-free, with exactly k simple negative zeros inside. The complete correction is uniformly smaller on that whole circle, not merely at its negative point. Rouché therefore gives the reciprocal count and sign in (4.7).

In the critical range, the direct limit is the macroscopic Beta kernel. Its signed-weight representation has positive initial mass, upward jumps, and negative variation below one half, so the same cosine-transform rectangle argument gives only simple negative zeros and zero-free counting circles. The reciprocal limit remains (4.8b). The local-uniform transfers at both circles are uniform when $q, n - rq, nh, n/h$ exceed their finite gate. The explicit Hankel, Beta, and reciprocal boundary estimates are proof-dossier D4.

For the degree, every interior point must have dilation at least both $\lceil n/r \rceil$ and $\lceil (n - q + 1)/(r - 1) \rceil = c$. At dilation c , put one in every coordinate and distribute the additional b units among the q coordinates outside H_0 , never exceeding $c - 2$ extra units per coordinate. The capacity inequality $q(c - 2) \geq b$ holds for large n . Then $x(H_0) = n - q < (r - 1)c$; every other effective hyperplane has size at most $q + r - 2$, and hence has sum at most $q + r - 2 + b < (r - 1)c$. This is a relative interior lattice point, proving the codegree and degree assertions. The phase labels in (4.8) follow by expanding the exact saddle equation (3.18); notably the finite $\pi/(r - 1)$ -offset is retained. $\square$

Proposition 4.4 (fixed-complement theorem).

Fix integers $r \geq 3$ and $q \geq 2$. For all sufficiently large n , every connected rank- r paving matroid having an effective hyperplane of size $n - q$ has only simple strictly negative h^ -zeros, with no restriction on the remaining legal effective hyperplanes.*

Proof. Put $k = r - 1$, $m = q - 1$, and $N = n - q + 1$. For the one-hyperplane model, split the outside-coordinate sum into the intervals $s = (b + 1)t + y$, $0 \leq y \leq t$, and inclusion-exclude the upper bounds. The outside fibre is

$$F_b(t, y) = \sum_{v=0}^{b+1} (-1)^v \binom{q}{v} \binom{(b+1-v)t + y + q - 1 - v}{q-1}. \quad (4.9)$$

Expanding in $\binom{y}{j}$, the required uniform-slice moments are

$$\begin{aligned} M_{a,N,j}(t) &= \sum_{i=0}^{a-1} (-1)^i \binom{N-1}{i} \binom{(a-i)t + N - 1 - i}{N+j-1} \\ &\quad - \sum_{i=0}^{a-2} \sum_{\ell=0}^j (-1)^i \binom{N-1}{i} \binom{t+1}{\ell} \binom{(a-i-1)t + N - 2 - i}{N+j-\ell-1}. \end{aligned} \quad (4.10)$$

Equations (4.9)–(4.10), with the common-endpoint subtraction, are an exact finite expression for the Ehrhart polynomial. Its highest-slope part is obtained from the uniform rank- k numerator by m operators

$$T_{w,M}(f) = wx(1-x)f' + [1 + (wM - 1)x]f, \quad (4.11)$$

where the parameters are positive and converge to $1, 1/2, \dots, 1/m$. The remaining terms have smaller slope and the signed size bounds recorded in proof-dossier D3. On the middle annulus $Y/N \leq y \leq N/Y$, root-of-unity filtering gives a conjugate leading pair $2\mathcal{M} \cos \Phi/k$ and

$$\left| \frac{E}{\mathcal{M}} \right| + \left| \partial_\Phi \frac{E}{\mathcal{M}} \right| \leq C_{r,q} e^{-c_{r,q} Y} + o_N(1). \quad (4.12)$$

At the two endpoints, after the scalings $x = z/N^k$ and reciprocal normalization, the error and its first two derivatives tend to zero on compact sets. The limiting numerators are obtained from the uniform kernels by the positive operators (4.11), and their negative roots are simple. Thus the homotopy from the leading model to the exact one-hyperplane polynomial has no negative multiple root in any of the three regions. Its constant and leading coefficients stay positive, so roots cannot escape through zero or infinity; all roots remain simple and negative.

For $r = 3$, the same finite moment formula holds, but (4.12) is replaced by the two-saddle identity

$$J_{0,N}(-y^2) = \mathcal{M}_N(y) \cos \Phi_N(y), \quad \Phi'_N(y) \geq cN/(1 + y^2), \quad (4.13)$$

and the direct and reciprocal limits are respectively products of $1 + (z d/dz)/j$ applied to $\cosh \sqrt{z}$, and the two parity kernels $\cosh \sqrt{z}$ and $\sinh(\sqrt{z})/\sqrt{z}$. Their function and derivative never vanish simultaneously on the negative axis, so the same three-region homotopy closes rank three.

Finally, every additional hyperplane meets the distinguished one in at most $r - 2$ elements. Because q is fixed, packing gives only $O_{r,q}(n^{r-3})$ such corrections; after the normalizations above their coefficients satisfy the same strict output-box bounds as the error in (4.12). They therefore enter the same homotopy without

changing the argument. The full derivation of (4.9)–(4.13), including the endpoint two-jet estimates, is proof-dossier D3. $\square$

Proposition 4.5 (rank-three giant-hyperplane theorem).

There is an integer N_{giant} such that, for every $n \geq N_{\text{giant}}$, every connected rank-three paving matroid whose largest effective hyperplane has size $s \geq \lceil 2n/3 \rceil - 1$ has only simple strictly negative h^ -zeros. All legal attached hyperplanes are included.*

Proof. Put $q = n - s$, $N = s + 1$, and $m = q - 1$. The one-hyperplane numerator has a simple negative spectrum by the exact rank-two slice representation. Write the complete numerator as

$$p_\tau = p_{\text{single}} - \tau \Delta, \quad 0 \leq \tau \leq 1. \quad (4.14)$$

where Δ is the sum of all legal attached-hyperplane corrections. If $m/N \rightarrow 0$, the sublinear-complement theorem recorded as **S9** together with its dependency revision **S9R1** in proof-dossier D3 applies uniformly; it covers both fixed and divergent q . It therefore remains to work on an arbitrary compact interval of positive values of $m/N \subset (0, 1/2]$.

On that compact interval, root-of-unity filtering writes

$$p_{\text{single}} = J + R_{\text{single}}, \quad J(-y^2) = \mathcal{M} \cos \Phi, \quad \Phi' > 0. \quad (4.14a)$$

The exact parent-remainder, packing, and contour estimates give, for a fixed sufficiently large threshold B ,

$$\frac{|R_{\text{single}} - \tau \Delta|}{\mathcal{M}} < \frac{1}{2}, \quad 0 \leq \tau \leq 1, \quad (4.15)$$

at every phase-grid point with $ny^{2/3} \geq B$. The estimate is uniform in the compact ratio interval and in every attached family; its proof combines the parent remainder with the full attached correction and splits the zero, finite-positive, and far-end scales, as recorded in proof-dossier D3. On the remaining compact endpoint, the weighted kernel of (4.14) has initial value one, downward variation below one third, and only upward jumps. Consequently the function and derivative are uniformly separated from simultaneous zero for every $\tau \in [0, 1]$.

The constant term, degree, and positive leading coefficient of (4.14) are independent of τ . A root therefore cannot escape through zero or infinity. At the first parameter where the simple negative spectrum could fail, a negative multiple root would occur. The compact endpoint separation excludes it below B , and (4.15), together with $\Phi' > 0$, excludes it above B . Thus the homotopy reaches the complete polynomial at $\tau = 1$ with all roots simple and negative. A failure sequence and compactness of $m/N \in [0, 1/2]$ convert the sequential statement into one common threshold. $\square$

5 Fixed-rank exhaustion

We now assemble the fixed-rank result. The following three lemmas are the uniform regime statements supplied by Sections 3–4 and the endpoint kernels.

Lemma 5.1 (bounded critical excess).

Fix $r \geq 4$ and $C \geq 0$. Let

$$D_u = n - \left\lceil \frac{n}{r} \right\rceil, \quad s = \begin{cases} \max_{H \in \mathcal{H}(M)} |H|, & \mathcal{H}(M) \neq \emptyset, \\ r - 1, & \mathcal{H}(M) = \emptyset. \end{cases}$$

For all sufficiently large n , every connected rank- r paving matroid with $s \leq D_u + C$ has a simple strictly negative h^ -spectrum.*

Proof. There are two different endpoint problems. If $s \leq D_u - 1$, put $g = D_u - s \geq 1$. This is the infinite family $F_{r,b,g}$, not a finite collection. Fix sufficiently large endpoint counts j, k . If the transfer were not uniform over all g , choose a failure sequence. After fixing the finite residue b , either g is constant on a subsequence or $g \rightarrow \infty$; Proposition 4.1 then gives the limiting kernel $F_{r,b,g}$ or $V_{r,b}$, respectively, on the same zero-free counting circle. Rouché's theorem contradicts failure. The direct endpoint has the same failure-sequence compactness, so both endpoint blocks have common thresholds without taking a maximum over all g .

If $\delta = s - D_u \in \{0, \dots, C\}$, write

$$\begin{aligned} b_u &= r \lceil n/r \rceil - n, \\ \ell &= \left\lceil \frac{\delta + 1 - b_u}{r - 1} \right\rceil, \\ c &= \lceil n/r \rceil + \ell, \quad b = b_u + r\ell, \quad a = (r - 1)\ell + b_u - \delta - 1. \end{aligned} \tag{5.1}$$

For fixed r, C , only finitely many (b, a) occur, and the $E_{r,b,a}$ -kernels in Proposition 4.1 give common endpoint thresholds. In the first subdomain, the phase grid (2.12) is ordered and Corollary 3.3 gives alternating strict signs. In the bounded-excess subdomain the required signs instead come from Proposition 3.3A. The intrinsic degree is

$$n - \max \left\{ \left\lceil \frac{n}{r} \right\rceil, \left\lceil \frac{s + 1}{r - 1} \right\rceil \right\}; \tag{5.2}$$

Indeed, in the first subdomain take $c = \lceil n/r \rceil$ and $b = rc - n$. At dilation c , assign one to every coordinate and add one to any b coordinates. Every effective hyperplane then has sum at most

$$s + b \leq D_u - 1 + b = (r - 1)c - 1,$$

so this is a relative interior lattice point. In the bounded-excess domain, use the parameters in (5.1). The largest hyperplane is unique for large n , and the exact identities give $q \geq b$; assign the b extra units to distinct coordinates outside it. Its sum is $s = (r - 1)c - a - 1 < (r - 1)c$. Every other effective hyperplane has size at most $q + r - 2$, and its total is at most $q + r - 2 + b = c + a + r - 1 < (r - 1)c$. Conversely, positivity of all coordinates and the largest-hyperplane inequality force both lower bounds in (5.2). Thus the codegree and degree are exact. The endpoint counts plus the intermediate sign changes equal this degree. Local Rouché disks at the endpoints and strict sign separation in the middle also exclude multiplicity. $\square$

Lemma 5.2 (subcritical giant hyperplane).

Fix $r \geq 5$ and $0 < p_{\max} < 1/r$. For all sufficiently large n , every connected rank- r paving matroid whose largest effective hyperplane has complement size q satisfying

$$2 \leq q \leq p_{\max} n \tag{5.3}$$

has a simple strictly negative h^* -spectrum.

Proof. Choose $Q = Q(r, p_{\max})$ from the uniform divergent-complement estimate. For $Q \leq q \leq p_{\max} n$, distinct effective hyperplanes meet in at most $r - 2$ elements, so every other hyperplane has size at most $q + r - 2$. Proposition 4.3 treats both endpoint blocks and gives the exact degree. The complete middle sign is Proposition 3.4, not Proposition 3.1: the largest hyperplane lies beyond the latter proposition's central-density domain. The paving intersection bound places every other hyperplane in the separated error term already included in Proposition 3.4. Ordered first hits of the levels between the two endpoint phases yield every remaining root.

Only the finite set $q = 2, \dots, Q - 1$ remains. For each fixed q , the fixed-complement theorem, Proposition 4.4, supplies a threshold N_q , including all legally attached hyperplanes. Taking the maximum of these finitely many thresholds and the uniform $q \geq Q$ threshold proves the lemma. Equivalently, any failing sequence has either a divergent- q subsequence or a constant- q subsequence, and both are excluded. $\square$

Lemma 5.3 (critical moving giant hyperplane).

Fix $r \geq 5$. Suppose a sequence of connected rank- r paving matroids has largest-hyperplane complement q with

$$\frac{q}{n} \longrightarrow \frac{1}{r}, \quad n - rq \longrightarrow \infty. \quad (5.4)$$

Then its h^* -polynomials eventually have only simple strictly negative zeros.

Proof. Fix large integers J, k before sending $n \rightarrow \infty$. At the direct endpoint, the disk ending at $x_L = -R_{p,J}/n^r$ contains exactly J simple negative roots and has boundary sign $(-1)^J$. At the reciprocal endpoint, after fixing the finite residue parameter, the exterior of $x_U = -\Lambda/T_{a,k}^d$ contains exactly k simple negative roots and has boundary sign $(-1)^{D-k}$.

The four quantities

$$q, \quad n - rq, \quad nh, \quad n/h \quad (5.5)$$

exceed a common fixed-rank threshold throughout the interval between these endpoint boundaries. Here h is the increasing phase-scale parameter: along the leading branch,

$$K = \frac{a_0 h}{dp} \quad \text{and} \quad a_s = \frac{a_0 t^r}{1 - p} = h^{r-1} \frac{\sin(r\theta)}{\sin \theta}$$

are strictly increasing with $\theta \in (\pi/r, \pi/(r-1))$; the saddle equation then makes h strictly increasing with t . The first two quantities tend to infinity by (5.4), and the latter two are made large by choosing J, k first. Proposition 3.4 then gives the complete actual sign at every intermediate U_n -phase level. Taking the first point at which each level $(J+1)\pi, \dots, (D-k-1)\pi$ is reached gives an ordered grid without assuming a stronger global monotonicity statement. The endpoints and grid supply $J + (D - J - k) + k = D$ distinct negative roots, so degree exhaustion makes them all simple. Finite residue classes are handled by a finite subsequence maximum. $\square$

Lemma 5.4 (the complete rank-three class).

There is an integer N_3 such that every connected rank-three paving matroid on at least N_3 elements has a simple strictly negative h^* -spectrum.

Proof. For $b \in \{0, 1, 2\}$ and $g \geq 1$, its reciprocal kernels are

$$F_{b,g}(z) = \sum_{j \geq 0} \Pr\{\text{Bin}(3j + b, 2/3) \leq 2j + b + g - 1\} \frac{z^j}{(3j + b)!}. \quad (5.6)$$

At $T_{i,b} = 2\pi(i + b/3)/\sqrt{3}$, exact negative-axis signs give $(-1)^i F_{b,g}(-T_{i,b}^3) > 0$, uniformly in g . Full complex-circle counts give exactly i roots inside $|z| = T_{i,b}^3$; hence all zeros of every $F_{b,g}$ are simple and negative.

For completeness, the sign is not a numerical extrapolation. Writing the kernel as the three-pole sum plus its fixed-radius remainder, the normalized remainder $Q_R(T)$ and the positive comparison $d(T)$ satisfy

$$Q_R(T) < d(T) \quad (T \geq 3). \quad (5.6a)$$

On $[3, 10^6]$, (5.6a) follows from 36 adjacent rational intervals: for each fixed rational radius, Q_R is convex and d is concave, so the two endpoint inequalities certify the whole interval. For $T \geq 10^6$, the analytic tail bound is $(Q_R/d < 0.975)$. Since every $T_{i,b} > 3$, this proves all signs. The independent full-circle estimate has the same dominant three-pole sum and a strict uniform boundary margin; the argument principle therefore gives exactly i roots. The rational interval certificate and the infinite-tail inequality are included in proof-dossier D3.

The direct endpoint limits are indexed by legal macroscopic spectra $\alpha = (\alpha_j)$, $\sum \alpha_j \leq 1$, $\max \alpha_j < 1$:

$$F_{\alpha}^{(3)}(z) = \sum_{\ell \geq 0} \left[1 - \sum_j \Pr\{\text{Bin}(3\ell, \alpha_j) \geq 2\ell + 1\} \right] \frac{z^\ell}{(3\ell)!}. \quad (5.6b)$$

Every function in (5.6b) has infinitely many simple negative zeros. The proof splits at the largest two masses. Small spectra satisfy a direct Beta–Rouché bound; otherwise the common signed weight has downward variation

$$V_- = \sum_{j \geq 3} \frac{\alpha_j}{2 - 3\alpha_j} + 1 - g(\alpha_2) < \frac{1}{2}, \quad (5.6c)$$

where, if $p_b \in (2/3, 1)$ is determined by $b^2(1 - b) = p_b^2(1 - p_b)$,

$$g(b) = \frac{p_b}{3p_b - 2} - \frac{b}{2 - 3b}. \quad (5.6c')$$

The three possible ranges of α_2 reduce to the explicit Bernstein-positive rational inequalities in proof-dossier D3. Upward jumps, finite total variation, and the gamma multiplier then give the asserted simple negative spectrum. This also pays countable spectra by dominated truncation.

Suppose a common threshold fails and choose a failing sequence. If $s \geq \lceil 2n/3 \rceil - 1$ along an infinite subsequence, Proposition 4.5 excludes that subsequence; this includes both fixed and divergent complements of the largest hyperplane. We may therefore pass to a subsequence with $s < \lceil 2n/3 \rceil - 1$. If $s/n \leq a < 2/3$ for some fixed a , packing gives a limiting spectrum for (5.6b); the exact contour formula converges uniformly on the direct circle, while the same size-layer estimate gives

$$\frac{|E|}{R^n} \leq C_a(1 + t^3)e^{-c_a t} \quad (5.6d)$$

on the moving middle axis. Thus the direct roots from (5.6b), the alternating middle grid, and the reciprocal kernel roots transfer uniformly.

The only remaining possibility is $s/n \rightarrow 2/3$ from below. Put $c = \lceil n/3 \rceil$, $D = n - c$, $b = 3c - n$, and $g = D - s$. If $g \rightarrow \infty$, the reciprocal limit is $V_{3,b}$; if g is bounded, fix the finite pair (b, g) and use $F_{b,g}$. The zero-free circles proved above transfer the exact reciprocal block. Independently, compactness of the hyperplane densities gives a direct limit (5.6b), so a fixed direct block also transfers. Between them, the exact double-moving formula satisfies (5.6d) in the separated part and the complementary-error estimate (3.22c) in the bounded critical window. Hence every integer phase has its alternating complete-polynomial sign. Ordered first hits give (D-j-k) middle roots; together with the (j) direct and (k) reciprocal roots, they exhaust the actual degree (D). All roots are therefore simple and negative, contradicting the chosen sequence. $\square$

Lemma 5.5 (the complete rank-four class).

There is an integer N_4 such that every connected rank-four paving matroid on at least N_4 elements has a simple strictly negative h^ -spectrum.*

Proof. Let $D_u = n - \lceil n/4 \rceil$, $s = s(M)$, and $q = n - s$. The exact regime theorems needed here are:

$$\begin{array}{l|l} I_0 & s \leq D_u : \text{Lemma 5.1 with } C = 0, \\ I_1 & D_u < s \leq D_u + C : \text{Lemma 5.1 for fixed } C, \\ I_2 & q/n \rightarrow 0 : \text{fixed } q \text{ or } q \rightarrow \infty \text{ subcritical}, \\ I_3 & q/n \rightarrow p_0 \in (0, 1/4) : \text{subcritical giant profile}, \\ I_4 & q/n \rightarrow 1/4, n - 4q \rightarrow \infty : \text{critical giant profile}. \end{array} \quad (5.7)$$

In I_2 , a bounded q subsequence is fixed and uses the exact fixed-complement theorem, Proposition 4.4; a divergent q subsequence uses Propositions 3.5 and 4.3. Regimes I_3 and I_4 use the corresponding interior and moving-critical clauses of Proposition 3.5 together with the endpoint counts of Proposition 4.3. In every case, the endpoint circles contain exact fixed root counts, the appropriate phase grid has complete-polynomial signs, and the codegree construction in Proposition 4.3 gives the actual degree; ordered first hits therefore exhaust all roots.

It remains only to see that this finite regime list is exhaustive. Put $c = \lceil (s+1)/3 \rceil$, $a_n = 3c - s - 1 \in \{0, 1, 2\}$, and $b = 4c - n$. If b is bounded along a failing subsequence, then $s - D_u$ is bounded and I_1 applies. If $b \rightarrow \infty$, then $n - 4q = 3b - 4a_n - 4 \rightarrow \infty$, and a subsequential limit of $q/n \in [0, 1/4]$ lies in exactly one of I_2, I_3, I_4 . Hence no failing rank-four sequence exists. $\square$

Proof of Theorem 1.1. First assume M is connected. Ranks three and four are Lemmas 5.4 and 5.5.

Now fix $r \geq 5$, and suppose no common threshold exists. Choose a failing sequence with $n \rightarrow \infty$. For each member write $s = s(M)$ and $D_u = n - \lceil n/r \rceil$, in accordance with (1.0).

If $s \leq D_u$ infinitely often, Lemma 5.1 with $C = 0$ is a contradiction. Otherwise put $\delta = s - D_u > 0$. If δ is bounded on a subsequence, Lemma 5.1 with one fixed bound C is again a contradiction. Thus we may assume $\delta \rightarrow \infty$. With $q = n - s$ and $b_u = r \lceil n/r \rceil - n \in \{0, \dots, r - 1\}$,

$$n - rq = r\delta - b_u \rightarrow \infty. \quad (5.8)$$

After taking a subsequence, $q/n \rightarrow p_0 \in [0, 1/r]$. If $p_0 < 1/r$, choose one fixed $p_{\max} \in (p_0, 1/r)$ and apply Lemma 5.2. If $p_0 = 1/r$, apply Lemma 5.3 using (5.8). Every possible bad sequence has been contradicted, so there is a common connected threshold.

It remains to classify disconnected paving matroids for fixed r and $n > r$. A component containing a circuit has rank at least $r - 1$, and $2(r - 1) > r$, so there is at most one such component. Every other connected component is a one-element coloop. Disconnectedness and the total rank force

$$M = U_{r-1, n-1} \oplus U_{1,1}. \quad (5.9)$$

Deleting the fixed coloop coordinate is an integral affine isomorphism of base polytopes and preserves the Ehrhart polynomial. The fixed-rank uniform hypersimplex theorem for rank $r - 1$ supplies a threshold for (5.9) when $r - 1 \geq 3$, by Adiprasito–Zhang [1, Theorem 3.1]. For $r = 3$, put $m = n - 1$. The rank-two numerator formula [2] is $P_m(x) = \sum_j \binom{m}{2j} x^j - mx$. At $x = -\tan^2 \theta$, $0 \leq \theta < \pi/2$, it gives

$$\cos^m \theta P_m(-\tan^2 \theta) = \cos(m\theta) + m \sin^2 \theta \cos^{m-2} \theta.$$

For $m \geq 5$, the nonnegative second term has maximum $2(1 - 2/m)^{(m-2)/2} < 1$: the maximum decreases with m , and its square at $m = 5$ is $108/125 < 1$. Hence the values at $\theta = j\pi/m$ have sign $(-1)^j$. If m is odd, this gives $\lfloor m/2 \rfloor$ disjoint sign-changing intervals. If m is even, the endpoint limit has sign $(-1)^{m/2}$ and supplies the last interval. Since $\deg P_m = \lfloor m/2 \rfloor$, all roots are distinct and strictly negative. Thus $n \geq 6$ suffices for this disconnected rank-three subcase; simplicity is justified by this argument, not by an unstated simplicity assertion in [2]. Taking the maximum of the connected and disconnected thresholds proves the theorem. $\square$

6 The growing-rank root-density theorem

We prove Theorem 1.2. Suppress the index ν , and define

$$J = \lceil r^7 A^4 \rceil, \quad K = \lceil r^5 A^4 \rceil. \quad (6.1)$$

Condition (1.1) implies

$$X \rightarrow \infty, \quad \frac{r^4}{n} \rightarrow 0, \quad \log n \gg r^4 A^4 \gg r^3 \log(2r). \quad (6.2)$$

At the J -th direct-end grid point,

$$(n - 1)t_J = rJ(1 + o(1)) = r^8 A^4(1 + o(1)). \quad (6.3)$$

At t_J , the first two quantities in Proposition 3.0 are paid by (6.2)–(6.3). On $[t_J, t_-]$, $\kappa = (n-1)t$ increases, whereas the logarithmic-separation and the two factors containing t are weakest at the right endpoint. There,

$$\log[(C_1 r)^r C_*^r t_- \log(1 + \kappa)] \leq -\frac{\log n}{K_0 r} + O(r \log(2r) + A) \longrightarrow -\infty, \quad (6.3a)$$

and the same estimate without $(C_1 r)^r$ is stronger. Moreover $\log(1/t_-) = \log n/(K_0 r)$ dominates $r^2 + r \log(2r)$ by (6.2). Since $t \log(1 + (n-1)t)$ increases with t , these right-end estimates and the left-end κ -estimates pay all five gates throughout $[t_J, t_-]$. Proposition 3.0, not merely a one-hyperplane central limit, gives the complete h_M^* -sign on this ultra-small interval. Proposition 3.1 then covers $[t_-, 1/4]$, $[1/4, 4]$, and the large- t interval down to the reciprocal K -scale. Corollary 3.3 gives

$$(-1)^j h_M^*(-t_j^r) > 0, \quad J \leq j \leq D - K. \quad (6.4)$$

Thus the intervals between consecutive grid points contain

$$D - K - J \quad (6.5)$$

distinct strictly negative zeros.

By Proposition 4.2, the reciprocal endpoint contributes $K - 1$ additional simple strictly negative zeros beyond its boundary radius. One last zero is forced between that boundary and the last phase-grid point. Indeed, if $b = r \lceil n/r \rceil - n$, the phase expansion (2.14) gives

$$\frac{n}{T_{K,b}} < t_{D-K} < \frac{n}{T_{K-1,b}}, \quad (6.6)$$

while the reciprocal boundary sign is

$$\text{sign } h_M^*(-n^r/T_{K-1,b}^r) = (-1)^{D+K-1}, \quad (6.7)$$

opposite to the grid sign $(-1)^{D-K} = (-1)^{D+K}$. The reciprocal block therefore contributes K zeros disjoint from (6.5). The total is $D - J$.

It remains to justify the degree. Put $c = \lceil n/r \rceil$ and $b = rc - n \in \{0, \dots, r-1\}$. Every interior lattice point of $kP(M)$ has all coordinates at least one and coordinate sum rk , so necessarily $k \geq c$. At $k = c$, start with all n coordinates equal to one and increase any b coordinates to two. Since (1.1) implies $c > 2$ eventually, all coordinate inequalities are strict. For every effective hyperplane,

$$x(H) \leq |H| + b \leq D - 1 + b = (r-1)c - 1 < (r-1)c, \quad (6.8)$$

so every hyperplane inequality is also strict; lower-rank flat inequalities follow from paving and the coordinate bounds. Thus the codegree is exactly c . Since connectivity gives $\dim P(M) = n - 1$,

$$\deg h_M^* = n - \left\lceil \frac{n}{r} \right\rceil = D. \quad (6.9)$$

Finally, (1.1) and (6.2) imply $J/D \rightarrow 0$. This proves Theorem 1.2. □

7 Scope, prior work, and limitations

The paving Ehrhart formula and the object-level hyperplane decomposition are prior work. Results for rank-two sparse paving matroids and the recent fixed-rank hypersimplex theorem are also prior work. In particular, the uniform subfamily of Theorem 1.1 is not claimed as new.

Adiprasito and Zhang [1, Theorem 3.1] prove that, for each fixed $k \geq 3$, the ordinary hypersimplex $\Delta(k, n)$ has a numerator with only distinct negative real zeros for all sufficiently large n . Their Theorem 1.1 gives

negative real-rootedness for rank three and $n \geq 6$, with simplicity asserted for $n \geq 32$. Theorem 1.1 here extends the object scope of their eventual fixed-rank theorem from uniform to all paving matroids; it does not improve their small-order rank-three range.

They also remark that their threshold can be quantified as $N(k) = k^{O(k^2)}$. With that growth bound, the uniform specialization in the regime (1.1) is already covered by their stronger full-root conclusion. Rank growth alone is therefore not the new part of Theorem 1.2: its increment is the nonuniform paving range and the explicit root-density window, with the stated hyperplane restriction. Theorem 1.2 is not a strengthening of their full-root theorem on uniform matroids.

The residue entire functions in [1, Section 3.3] and the phase-grid, endpoint-transfer, and degree-exhaustion architecture in its Sections 3.4–3.5 are prior analytic mechanisms used here. The full text of version v1 was checked; it contains no theorem covering all nonuniform paving matroids or the nonuniform conclusion of Theorem 1.2. This source comparison does not establish global priority over every other or unindexed work.

Theorem 1.1 supplies no explicit N_r and no threshold uniform in r . Theorem 1.2 assumes connectivity, (1.1), and the effective-hyperplane size bound; it leaves $o(D)$ zeros unclassified. Neither theorem resolves a general real-rootedness conjecture for arbitrary matroid base polytopes.

8 Reproducibility

The frozen prewrite package contains 216 source, proof, and certificate files. Their ordered path-and-SHA-256 manifest is

loops/PAVING-HSTAR-PUBLICATION-PREP-0001/A1-证明依赖闭包.sha256.

The finite certificates in the package check only explicitly stated rational inequalities and finite continuous covers. They do not replace the analytic proofs above. The manuscript and any public artifact must be rebuilt if a file in the frozen manifest changes.

For auditability, the principal manuscript assertions map to the frozen proof groups as follows; the manifest fixes the exact bytes of every named file and its bound revisions.

- Equations (2.1)–(2.9): S10, S10R1, and S11 in PAVING-HSTAR-HIGHER-RANK-CRITICAL-KERNEL-GATE-0001.
- Proposition 3.0: S17 and S20–S17门下的完整原端保号 in PAVING-HSTAR-CROSS-RANK-FACT-GATE-0001, together with that gate’s S13.
- Propositions 3.1–3.2: S18, S14, and S15 in PAVING-HSTAR-CROSS-RANK-FACT-GATE-0001, plus the full low-Euler estimate in S11.
- Proposition 3.3A: S59 and S91 in PAVING-HSTAR-HIGHER-RANK-CRITICAL-KERNEL-GATE-0001.
- Proposition 3.4: the full geometry, moving, actual-correction, endpoint, and critical-moving groups listed in proof-dossier D4.
- Proposition 3.5: S50–S56 in PAVING-HSTAR-HIGHER-RANK-CRITICAL-KERNEL-GATE-0001, as itemized in proof-dossier D5.
- Proposition 4.0: the complete macro-spectrum theorem in PAVING-HSTAR-TWO-MACRO-GATE-0001/S3, with its frozen analytic dependencies.
- Proposition 4.1 and Lemma 5.1: S90 and S91 in PAVING-HSTAR-HIGHER-RANK-CRITICAL-KERNEL-GATE-0001.
- Proposition 4.3: the Wright, reciprocal, phase-joining, and critical endpoint groups in proof-dossier D4.
- Proposition 4.4: S1 and S2 in PAVING-HSTAR-FIXED-OUTSIDE-GATE-0001.
- Proposition 4.5: S20-全部巨H依赖整类统一实际全轴候选 in PAVING-HSTAR-GIANT-H-DEPENDENCIES-GATE-0001, together with the giant- hyperplane moving and endpoint groups itemized in proof-dossier D4.

- Lemma 5.2: S8 in PAVING-HSTAR-GENERAL-RANK-GIANT-ENDPOINT-GATE-0001 and the fixed-complement theorem in PAVING-HSTAR-FIXED-OUTSIDE-GATE-0001.
- Lemma 5.3: S4 in PAVING-HSTAR-GENERAL-RANK-GIANT-CRITICAL-MOVING-GATE-0001.
- The rank-three and rank-four joins in Theorem 1.1: S22, S22R1, S23, and S24 in PAVING-HSTAR-MOVING-CRITICAL-GATE-0001, and S57 with its audited interfaces in PAVING-HSTAR-HIGHER-RANK-CRITICAL-KERNEL-GATE-0001.
- Proposition 4.2 and Section 6: S15, S19, and S20 in PAVING-HSTAR-CROSS-RANK-FACT-GATE-0001.

These are proof dependencies, not separate novelty claims. Novelty and source coverage are recorded outside the 216-file proof closure in the publication preparation package.

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